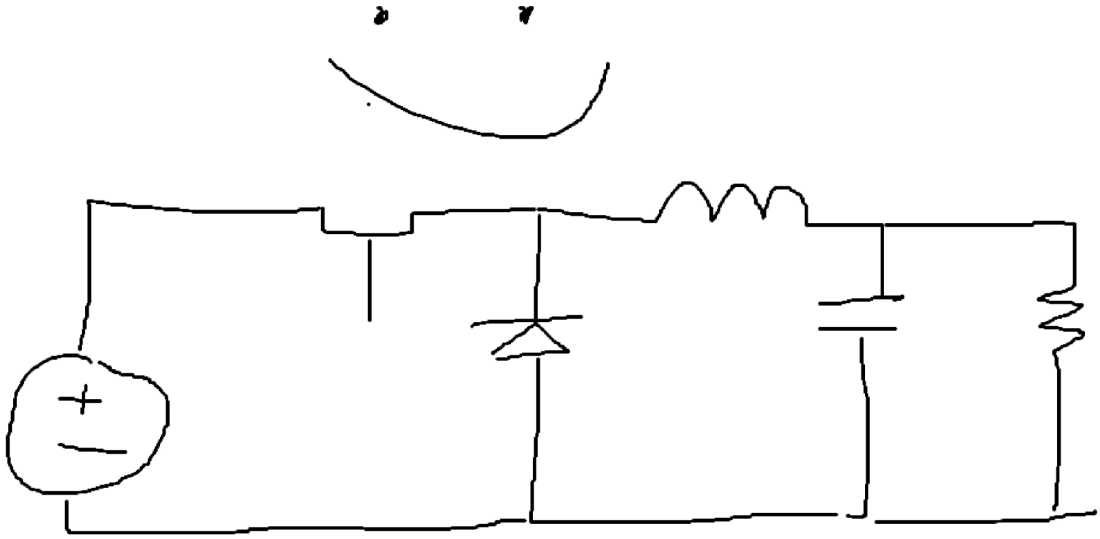
Formula Slug – Buck Converter Schematic Block
Power Electronics Onboarding Project 2026-2027

Overview

This project is centered around designing a buck converter to help power different boards on our LV system. This project requires you to design your own buck converter, sourcing components and creating the schematic and layout. You will also push it onto FS's github.



Our Low Voltage system consists of all our important logic and sensing boards, such as our Vehicle Control Unit, Telemetry boards, and Tractive Battery board. On all these boards, there are a variety of different components, who all may need to be powered by a different voltage level. For example, relays can operate at 12V, MCUs can operate at 5V, while some smaller chips run on as small as 3.3V. This is a big issue, because our low voltage bus runs on 12V!? If we overvolted those components, they would immediately stop working.

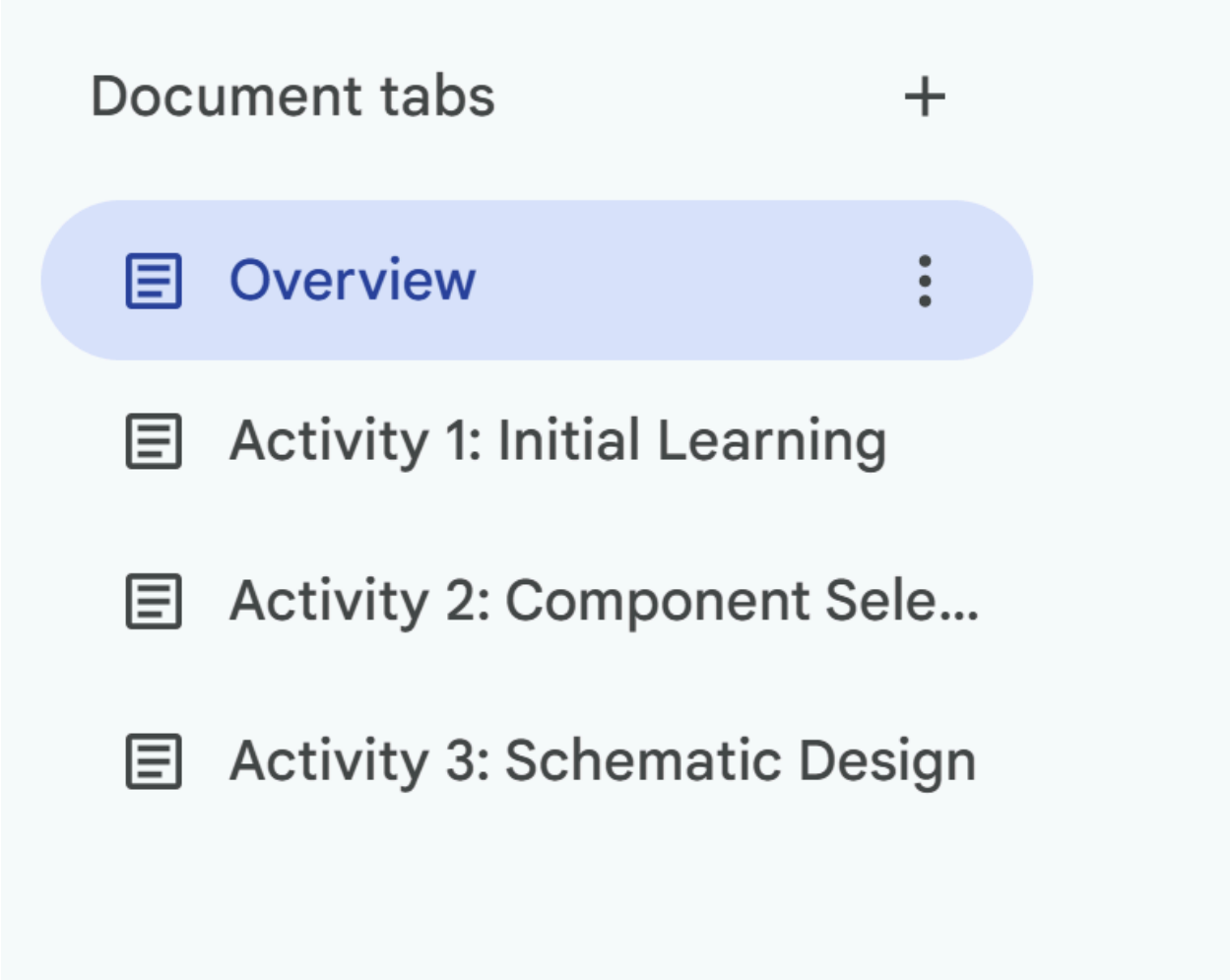
So what can we do to fix this?

A buck converter is used to transform a higher voltage to a lower voltage. We can use this to

power all of these components that require a much smaller power input, allowing it to properly function.

Navigating Project & Submission

There are 3 parts to this project: Initial Learning, Component Selection, and Schematic Layout. On the left side of the Google Doc, you will find the different tabs leading you to each activity.



Make a copy of this project, renaming it to "*Full Name* - Power Electronics Onboarding". Once every activity is completed, or before the due date, submit the document here. For example:

"Edmund Lai - Power Electronics Onboarding"

If there are any questions, message Edmund Lai on slack. There will be updates to this document if needed, check below for update logs.

Revision	Date	Updated Content
1.0	9/14/2026	Initial draft

Activity #1: Initial Learning

Listed below is a variety of initial resources that you can use if you are not familiar with electrical engineering concepts. I would suggest getting to know these well before you attempt to work on the buck converter. Any resource that talks about the same thing is good, as long as you understand the basics.

Analog Devices and Circuits:

1. Organic Chemistry Tutor: Ohm's Law
   https://www.youtube.com/watch?v=_rSHqvjDksg
2. Khan Academy, Current, Resistors, Ohm's Law:
   https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/circuits-part-1
3. Khan Academy, Compound Circuits:
   https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/electric-power
4. EEVBlog, Voltage Dividers:
   https://www.youtube.com/watch?v=xSRe_4TQbuo&list=PLvOlSehNtuHtWlH0UOZNtOn-FlFCn1GYw&index=2
5. EEVBlog, Kirchhoff's Laws:
   https://www.youtube.com/watch?v=WBfAEeEzDlg&list=PLvOlSehNtuHtWlH0UOZNtOn-FlFCn1GYw&index=13

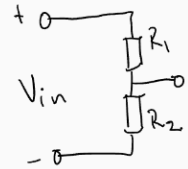
Listed below are more power related resources. This you will need to use to understand the theory behind how switch mode power supplies work.

1. MIT OpenCourseWave Power Electronics Course:
   a. Lecture 5: Intro to DC/DC, Part 1
2. Fundamentals of Power Electronics, by Robert W. Erickson and Dragan Maksimovic
3. Power Electronics by Daniel W. Hart

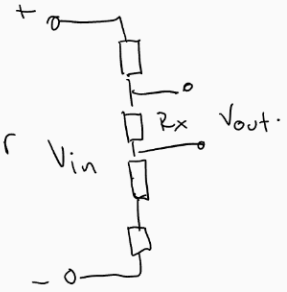
Please take good notes! Take notes in your notebook, here, whatever you would want to do. I would just like to see what you guys see and how you understand it!

Voltage dividers

→ Resistor V. D

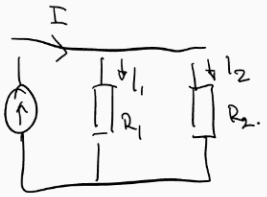


or



$$V_{out} = \frac{R_x}{R_{Total}} \cdot V_{in}. \quad ; R_x = \text{ref. resistor.}$$

→ Current. V. D



Use opp R

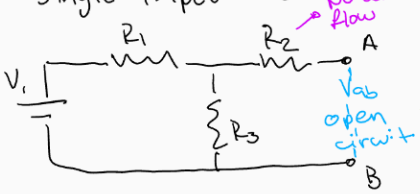
$$I_1 = \frac{R_2}{R_1 + R_2} \cdot I$$

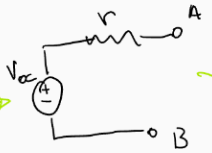
only works for 2 resistors.

Voltage Source

→ Thevenin's Theorem.

Any linear network that has V_s, I_s, R only can be reduced to a single source and single impedance





$$V_{TH} = V_{oc}$$

$$V_{oc} = \frac{V_1}{R_1 + R_3} \times R_3$$

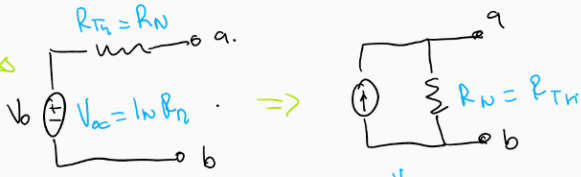
$$R_{Th} = \frac{V_o}{I_s} \qquad R_{Th} = R_2 + (R_1 \,//\, R_3)$$

$$= R_2 + \frac{R_1 R_3}{R_1 + R_3}$$

- To look for V_{Th}, remove the R_L (creating V_{oc}).
  V at the node A = V at the node between R_1 and R_3.
- To look for R_{Th}, remove V_s by shorting the points across it.

→ Norton's Theorem.

Any linear circuit containig V_s, I_s, R can be replaced by an equivalent circuit that consist of two components → an ideal current source in parallel with a resistor.



Th eq Circuit.

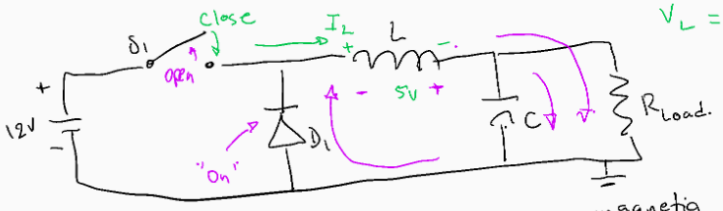
$$I_n = \frac{V_{oc}}{R_N}$$

- Find R_N, replace all Vs with short circuits and all Is with open ckts (nothing in that place)
  ↳ combine all $R_{1,2,3,\ldots}$ into 1.
- Find I_n, short the terminals a and b.
  ↳ use nodal analysis to find I_n.

Example to understand the general idea of a buck (step-down) converter.

Buck converter

→ Reduces the voltage at the input, while it increases the current.



$V_L = V_{out} - V_{in}$.
$12 - 5 = 7V$.

→ When S_1 → close, I_L increases → B (magnetic field) increases → L stores E
  ↳ charges the capacitor
  ↳ R_L receives I_{Bat}

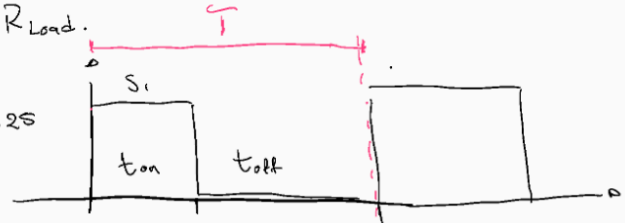
→ When S_1 → open, I_L decreases → B ↓ → this causes to induce v opposite polarity and becomes a current source.
  ↳ R_L receives I_L

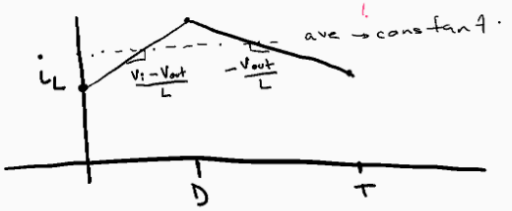
- If $V_L > V_C$, it can charge the capacitor
- if $V_L = V_C$, both L & C can discharge thru R_{Load}.

- V_{out} → V_i, Duty Cycle ("controlled, variable")

$D = \frac{t_{on}}{T}$ (T: Period) → $D = \frac{1ms}{4ms} = .25$ = 25%

↳ $V_{out} = V_i(D) = 12V(.25) = 3V$





Assuming: CCM and SS (average) (most be true)

$\frac{V_i - V_{out}}{L} DT = -\left(-\frac{V_{out}}{L}(1-D)T\right)$

$DV_i - DV_{out} = V_{out} - DV_{out}$

$V_{out} = V_i(D)$

- CCM = Continuos Conduction mode: L continuosly feeds i to the load. i_L never falls to zero during the entire switching cycle.
- SS (ave) – Steady state: value of the dynamic elements are constant.

* We can adjust Vout by adjusting the duty cycle.

Ex: 5A, 9V, Buck, D = 50%, Eff: 80%

$P_{in} = VI = 9 \cdot 5 = 45W$

$V_{out} = V_{in}(D) = 9(.50) = 4.5V$

$P_{out} = 36W$

$P_{out} = VI = I_{out} = \frac{P}{V} = \frac{36W}{4.5V} = 8A$

5A → 8A => i increase by a factor of 2.
9V → 4.5 => v decrease by a factor of 1.6

Calculate values for converter

Duty cycle: $V_{out} = V_{in}(D) \Rightarrow D = \frac{V_{out}}{V_{in}} = \frac{5V}{15} = 0.33 \rightarrow 33\%$ "if V_s fluctuates, V_o can be adjusted by changing D"

Inductance value: $L = \left(\frac{V_{in} - V_{out}}{\Delta i_L \, f_{sw}}\right) D = \frac{V_{out}(1-D)}{\Delta i_L \, f_{sw}} \Rightarrow L = \frac{(15-5)\times 0.33}{0.6 \times (5\times 10^5)} = 1.1 \times 10^{-5} H$

$= 11.1 \mu H$

Δi_L = L ripple current (A peak to peak)

$I_L = I_o = 2A$

$I_L = \frac{V_s D}{R_{load}}$

$R_{Load} = \frac{V_o D}{I_R} = \frac{V_o}{I_R} = \frac{5}{2A} = 2.5\Omega$

L_{min} for continuous current.

$L_{min} = \frac{(1-D)R}{2f} = \frac{(1-0.33)(2.5)}{2(5\times 10^5)} \approx 1.675 \mu H$

$r = 0.2$ to 0.4 ripple ratio

- lower r = ↓L but lower Δi_L
- higer r = ↑L but mor Δi_L

$\Delta i_L = 0.3 \times 2A = 0.6A$ target → 30%

Using 10uH: $\Delta L = \frac{(V_{in} - V_{out})D}{L \times f_{sw}}$

$\Delta i_L = \frac{(15-5)(0.33)}{(10\mu H)(5\times 10^5)} = 0.667A$

Ripple ratio = $\frac{\Delta L}{I_o}$

$\frac{0.667A}{2} \approx 0.33 \rightarrow 33\%$

Using 12uH

$\Delta i_L = \frac{(15-5)(0.33)}{(12\mu H)(5\times 10^5)} = 0.555A$

$\frac{0.555}{2} \approx 0.27 \rightarrow 28\%$ (Below target of 30%)

$L = 12\mu H$

10 uH | 12uH

Saturation i

$i_{peak} = i_{out} + \Delta i_L / 2$

$= 2A + 0.6/2 = 2.3A$

Capacitor

$C_{out} = \frac{\Delta i_L}{8 \times f_{sw} \times \Delta V_c} = \frac{0.555}{8 \times (5\times 10^5)(0.01)} = 1.387 \times 10^{-5}$

$13.9 \mu F$

Activity #2: Component Selection

After finishing Activity #1, you are ready to start selecting different components! When selecting components, please follow the LV Standards guideline to picking different form factors. Typically, our resistors and capacitors are both 0603, but this is just a general guideline. Typically do not go under 0603, but if we need a larger voltage rating, going up is completely fine. We typically only source from Digikey as well (Mouser if it is the only place available). We also do not want to use marketplace products. These are all filters that you can apply on Digikey, and more information is available in the LV stand ards document.



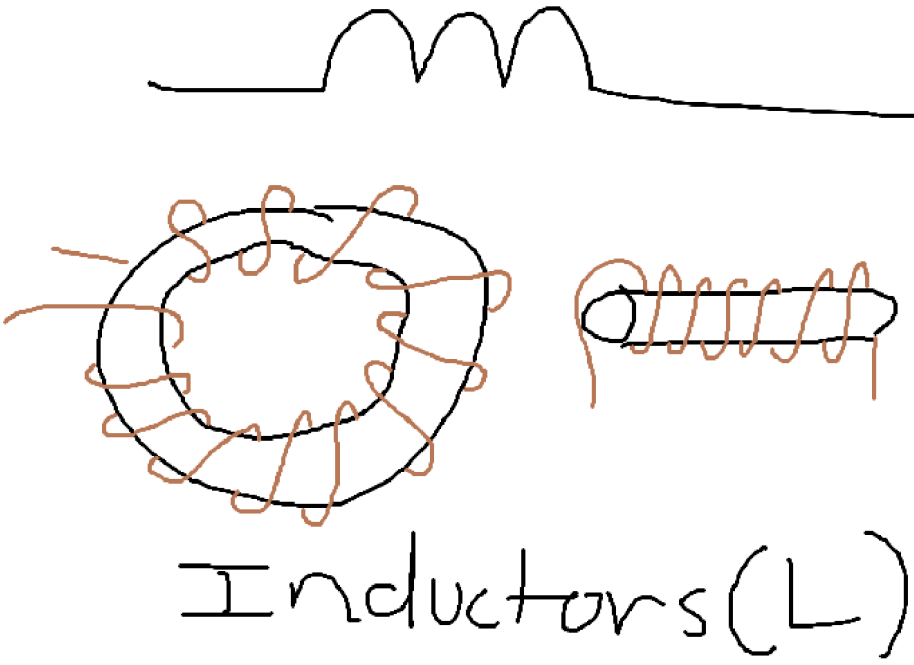
The first big thing to source is your Buck IC! This chip is used to adjust the duty cycle of the switches, and work with the feedback loop. For reference, here is the one we used last year but is now out of stock. The big thing is making sure it can output enough current to supply our boards! For FS5, we defaulted to a 2A buck, which worked perfectly fine (to my knowledge), so anything 2A and above is good! However, be sure to properly select this, because there are some tradeoffs that you may have to make if you oversize the buck converter too much. Feel free to ask me if you have any questions about that!

There are a couple I have in mind, but I would like you guys to do your own research and play with Digikey to be more familiar with the website! This is where we order almost all our parts, excluding some from Mouser. You will need to get familiar with it. Here are some initial parameters to get you all started on looking.

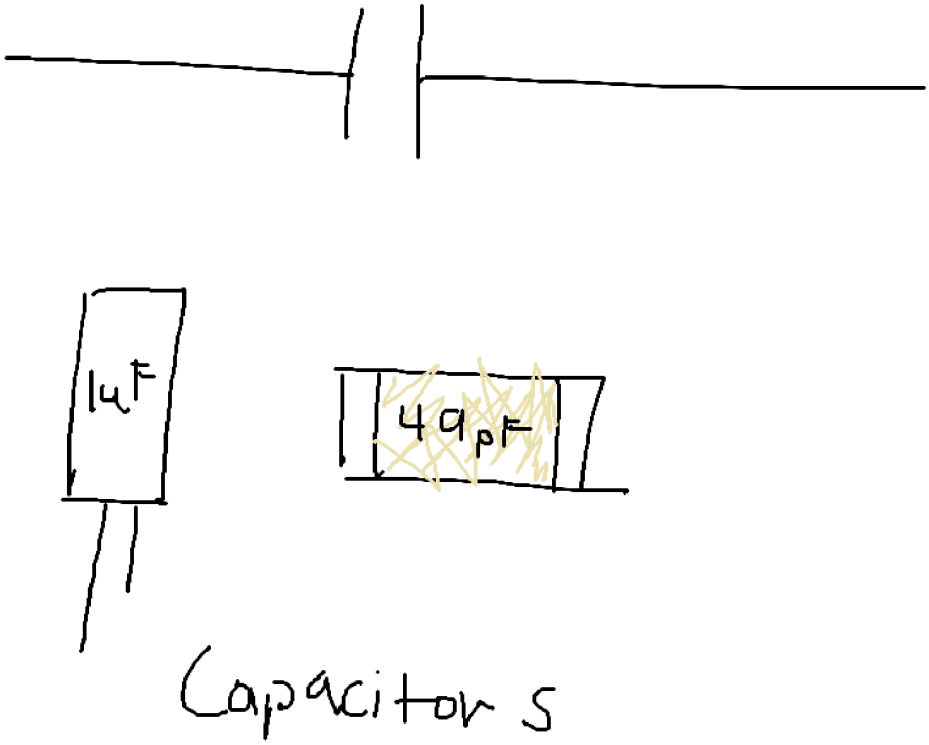
Input Voltage	~10-15V *
Output Voltage	5V
Output Current	Iout ≥ 2A
Switching Frequency	Fsw ≥ 500kHz**
Feedback Control	😳
Ripple Voltage	1-5%

*Though our LV bus is 12V nominally, it is actually 13.5V!
**with too low of a switching frequency, you upsize your inductor and output capacitor value, which could be an issue space wise. This is an arbitrary value I have set for the case of the onboarding project.

Whatever chip you decide to pick, be sure to properly read the entire thing, understanding what each pin does, what kind of internal structure it has, and the reasoning behind all of it.



One big thing you should check in an inductor is its saturation current. Make sure that your inductor does not oversaturate at the output current you are expecting, or else your inductor's inductance will drop so low that your output ripple will be extremely large.



Capacitors have a funny little thing called DC Biasing. Capacitors do not like staying at the specific capacitance they were designed to be at. Instead, whenever you apply any kind of voltage across one, their capacitance will drop by some percentage, typically shown in your datasheet. Make sure that your capacitances will be what you are expecting at your expected output voltage! A little hint: The output capacitors are there to help filter noise! There is something you have to do when picking your output capacitors to help reduce noise on the output end.



Make sure to try to minimize costs as much as possible! This is more of a good engineering practice to not just go with the best thing if it costs an unreasonable amount of money.

At the end of this activity, fill out this chart with the things that you have selected for ONE buck converter, with the costs per unit and size as well. If you need to add more rows, feel free to.

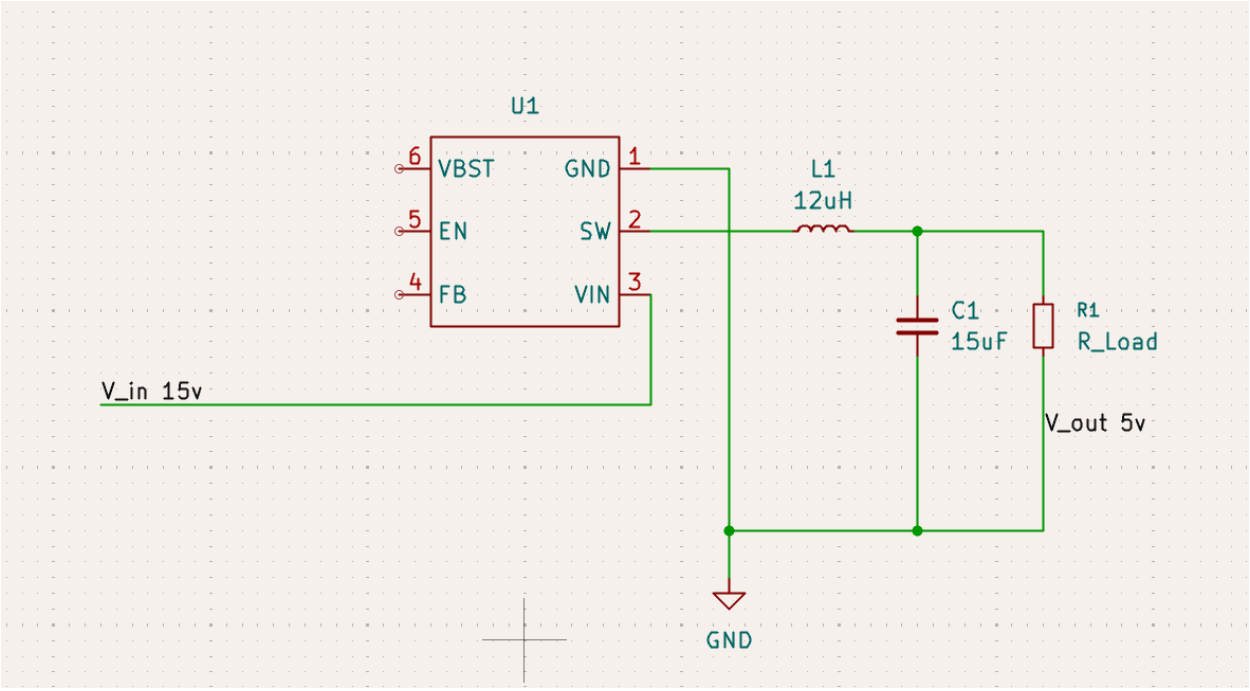
Component	Cost	Link	Units	Size
Buck IC	$1.22	https://www.digikey.com/en/products/detail/texas-instruments/TPS542025DDCR/28738814?s=N4IgTCBcDaICoAUDKBWALGADGFARXAwgEogC6AvkA		
Inductor	$0.47	https://www.digikey.com/en/products/detail/bourns-inc/SRN6045TA-120M/7315033		
Capacitor				
Total Cost				

Activity #3: Schematic and Layout Design

Here is where you start designing the schematic! Create a branch in the onboarding repo, and use the template KiCad project to make your buck converter. Be sure to follow the LV standards guide.

Please attend one of the KiCad workshops, where the electrical leads will teach you more about the software and what we expect.

Once you feel that your schematic and or layout is done, make a commit on the fs-electrical-onboarding repository, and push it onto github. You should make a commit no matter what, even if you did not finish completely. If you do not, you will not be considered for Power Electronics.



Extra Activity: Tractive System Shunt Current and/or Voltage Sense

If you happen to finish the buck converter, or you feel that the buck converter doesn't speak to you, you can also work on this project as well. This one will be in the form of a presentation / research project.

This project is based around adding to our current sensors within our Tractive Battery's container. Currently, we use a hall effect current sensor, which kind of looks like this:



For accuracy and data acquisition reasons, it would be nice to be able to add to this hall effect sensor with a shunt one, where you measure the voltage drop across a specially manufactured piece of metal with a specific resistance and voltage drop across, where you can then determine what current is going through it based on it. However, there are some constraints to this system, ie: isolation, heat dissipation, etc

Our current tractive battery current can go up to ~600A peak. We want to be able to measure up to this and down to the negatives as well in order to track regen current. Required by FSAE 26-27 Rules (EV.8.7.4), we need to be able to *prove* to judges that the Brake System Plausibility Device works through simulating tractive system current going through the current sensor. Find a way to be able to simulate this using a shunt if possible. At the moment, we have wire wrapped around a hall effect sensor multiple times, simulating higher currents with a smaller test current.

We would also like to be able to properly measure the voltage in our tractive system while we drive. We should find a way to take that voltage measurement, isolate it and pass it through to a

microcontroller so that we can have this data for the future.

When completing this research project, continue to write good documentation and make sure to optimize for costs when possible.

Have fun!